Fundamentals of Fog and Edge Computing

BCSE313L

WINTER 2024 - 2025

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PRP-217-4

Course Objectives

- To introduce IoT enabling technologies and its opportunities.
- To review underlying technologies, limitations, and challenges along with performance metrics and discuss generic conceptual framework in fog computing.
- To impart the knowledge to log the sensor data and to perform further data analytics.

Course Outcomes

- Explore technologies behind the communication and management of fog and edge resources.
- Learn the techniques for storage and computation in fogs, edges, 5G and clouds.
- Implement Internet of Everything (IoE) applications through fog computing architecture and use optimization techniques for the same.
- Analyze the performance and issues of the applications developed using fog and edge architecture.

Syllabus

Module:1	**Internet of Things (IoT) and New Computing Paradigms**	**6 Hours**
Introduction - Relevant Technologies - Fog and Edge Computing Completing the Cloud - Hierarchy of Fog and Edge Computing - Business Models – Edge Computing Platforms - Opportunities and Challenges		
Module:2	**Challenges in Federating Edge Resources**	**6 Hours**
Introduction - Methodology - Integrated C2F2T Literature by Modeling Technique - Integrated C2F2T Literature by Use - Case Scenarios - Integrated C2F2T Literature by Metrics – Threads - Standards		
Module:3	**Orchestration of Network Slices in Fog, Edge, and Clouds**	**6 Hours**
Introduction – Background - Network Slicing - Network Slicing in Software-Defined Clouds-Network Slicing Management in Edge and Fog - Internet of Vehicles (IoV): Architecture, Protocols and Seven-layer security model architecture for Internet of Vehicles - IoV: Network Models, Challenges and future aspects		
Module:4	**Optimization Problems in Fog and Edge Computing**	**6 Hours**
Preliminaries - The Case for Optimization in Fog Computing-Formal Modeling Framework for Fog Computing – Metrics - Further Quality Attributes - Optimization Opportunities along the Fog Architecture - Optimization Opportunities along the Service Life Cycle - Toward a Taxonomy of Optimization Problems in Fog Computing		

Syllabus

Module:5	**Middleware for Fog and Edge Computing**	**6 Hours**
Need for Fog and Edge Computing Middleware - Design Goals-State-of-the-Art Middleware Infrastructures - System Model - Case Study.		
Module:6	**Technologies in Fog Computing**	**7 Hours**
Fog Data Management - Smart Building - Predictive Analysis with FogTorch - Machine Learning in Fog Computing - Data Analytics in the Fog - Data Analytics in the Fog Architecture.		
Module:7	**Applications of Fog and Edge Computing**	**6 Hours**
Exploiting Fog Computing in Health Monitoring-Smart Surveillance Video Stream Processing at the Edge for Real - Time Human Objects Tracking-Fog Computing Model for Evolving Smart Transportation Applications - Testing Perspectives of Fog - Based IoT Applications - Legal Aspects of Operating IoT Applications in the Fog		
Module:8	**Contemporary Issues**	**2 Hours**

Books – Text and Reference

Text Book(s)	
1.	Buyya, Rajkumar, and Satish Narayana Srirama, Fog and Edge computing: Principles and Paradigms, 2019, 1st edition, John Wiley & Sons, USA.
Reference Books	
1.	Bahga, Arshdeep, and Vijay Madisetti, Cloud computing: A hands-on approach, 2014, 2nd edition, CreateSpace Independent Publishing Platform, USA.
2	OvidiuVermesan, Peter Friess, “Internet of Things –From Research and Innovation to Market Deployment”, 2014, 1st edition, River Publishers, India.

[Textbook Link]

Assessment

Assessment Head	Maximum Marks	Weightage	Tentative Date
Digital Assessment 1	20	10	**29-03-2025**
Quiz 1	10	10	**24-01-2025** (Moodle)
Quiz 2	10	10	**12-03-2025** (Moodle)
Continuous Assessment Test 1	50	15	As Per University
Continuous Assessment Test 2	50	15	As Per University
Total	**130**	**60**	